Aryaveer Singh

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Career Objective

Electronics and Communication undergraduate focused on securing Embedded Systems position to apply skills in microcontrollers, sensors, and circuit design, enhancing industry knowledge and facilitating hardware-software integration initiatives.

Skills

- Circuit Design
- PCB Design
- PCB Soldering
- Routing Protocols
- C Programming
- Python
- Machine Learning
- Arduino
- ESP32
- Internet of Things (IoT)
- Verilog
- Google Colab
- Effective Communication
- Interpersonal skills
- Blog

Work Experience

06/2025 - 07/2025

Instrumentation and Control Engineering Intern, **Reliance Industries**, Jamnagar

- Calibrated and commissioned field instruments, including RTDs and thermocouples, at Reliance Industries Limited.
- Employed dead-weight testers and multifunction calibrators for precise instrument calibration.
- Installed and troubleshot Bently Nevada 3500 monitoring systems and vibro-acoustic diagnostics.
- Captured and processed vibration data to evaluate alert and danger thresholds.

05/2025 - Current

Vice President, Avionics Club (2025-Present) :, **Avionics Club**

- Contributed innovative solutions to enhance team performance and outcomes.
- Utilized advanced technical skills to troubleshoot complex issues and implement effective solutions.
- Streamlined processes through various software and tools, optimizing overall performance.
- Demonstrated strong problem-solving abilities by resolving issues efficiently.
- Showcased PCB designing skills and trained club members in these techniques.
- Simulated critical circuitry of hardware wirings and models using simulations.

03/2025 - 04/2025

Project Trainee, **SKYLINE AEROSPACE**

- Company Overview: Project Trainee, Aircraft Design Technology - Skyline Aerospace (Remote)
- Completed a 1-month virtual internship focused on aircraft design fundamentals; contributed to project-based learning under the INSPACE-recognized program, engaging in technical discussions and team activities
- Project Trainee, Aircraft Design Technology - Skyline Aerospace (Remote)

07/2024 - Current

Event Management Head,, **Avionics Club**

- Directed planning and execution of technical events and hands-on drone/RC plane projects, managing team coordination and innovation-driven initiatives.
- Applied knowledge of applications, programming, and systems functionality to guide club members in addressing technical challenges during events.
- Assisted in conducting workshops focused on PCB design while developing a prototype for an in-house Flight Controller

10/2023 - 05/2024

Technical Subcomittee Member, **Avionics Club (Aeromodelling Club Of University)**

- Assisted in drone and RC plane initiatives within technical team, enhancing development, testing, and workshops over course of year-long tenure.

Education

Expected in 05/2027
B.Tech
Pandit Deendayal Energy University (PDEU), Gandhinagar

01/2023
Senior Secondary (XII), Science
Delhi Public School, Bopal

Trainings / Certifications

Drone Bootcamp Workshop: Completed a 2-day training on drone technologies at PDEU in collaboration with Hawaiadda Aerial Innovations, Pvt. Ltd.
Image Processing Workshop: I attended a 2-day workshop on image processing using Python, jointly organized by the Departments of Mathematics and Computer Engineering at PDEU
IoT Competition: Engaged in an IoT workshop and competition, demonstrating practical skills in embedded systems, and real-world IoT applications
MATLAB Hackathon: I participated in a MATLAB Hackathon to apply and enhance my data processing, and problem-solving skills
IoT-ML Workshop: Completed a workshop on IoT and Machine Learning, focusing on sensor integration, real-time data, and ML deployment using ESP32 and cloud platforms.
VLSI Workshop: Attended a certified workshop on VLSI Design using Verilog HDL, with hands-on experience in digital circuit design and testbench simulation.

Projects

- **IoT Wireless Pager System**
  Created an ESP32-based IoT pager for real-time alerts with cloud connectivity in time-critical scenarios.
- **Drone-Based Friend/Foe Detection System**
  Built a real-time object detection system using OpenCV to classify targets as Friend or Foe based on shape recognition.
- **Line-Following Mini Drone Prototype**
  Developed an autonomous mini drone with onboard vision and control for line tracking and stable flight.
- **GPS Tracker PCB module using STM-32**
  Designed and assembled a 4-layered GPS Tracker PCB using KiCad for reliable, compact, and efficient location tracking applications.
- **FM Transmitter and Receiver Hardware Module**
  Built an analog FM module on Vero board by soldering components, demonstrating basic modulation and reception.
- **Home power load automation system using ESP-32**
  Designed and deployed an IoT-based solution for automated control of home electrical loads, leveraging ESP-32 microcontrollers to enable wireless operation and smart scheduling of connected devices.

Extracurricular Activities

Scientific Volunteer, IPR(Institute of Plasma Research) DAE Platinum Jubilee Plasma Exhibition:

- *Served as a trained volunteer at IPR's national outreach event, engaging with visitors to explain plasma science concepts, fusion technology, and real-world applications through interactive exhibits and demonstrations.*

Accomplishments

- Got selected in the PoC Stage of **Robofest 5.0** in the '**Application Based Robot**' Category
- Secured 1st place in the **Mech-a-thon** competition conducted by within the university
- Secured 3rd place in a post-workshop **IoT** competition within the university during a 5-day IoT with Deep Learning workshop at PDEU